Ring Embedding in Faulty (n, k)-star Graphs

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Abstract

In this paper, we consider the ring embedding problem *in faulty (n, k)-star graphs. An (n, k)-star graph is recently proposed as an attractive interconnection network topology and also known as the generalized version of an n-star graph with scalability such that the number of nodes in the graph can be suitably adjustable by two dimensioning parameters n and k. Our scheme is proceeded in top-down style in such a manner that the resulting sub-stars maintain evenly distributed faults. We show that a ring of length $n!/(n-k)! - f$ can be found in an (n, k)-star graph having $n!/(n-k)!$ nodes when the number of faulty nodes f is at most $n-3$ and $n-k \geq 2$.*

1 Introduction

The underlying topology of multiprocessors system is modeled as an undirected graph where the nodes represent the processing elements and the edges represent the bi-directional communication channels. Embedding a guest(source) graph in a host(target) graph has long been used to model the processor allocation in the parallel and distributed computing environments; the *guest graph* represents a parallel or distributed algorithm and *host graph* does an interconnection network of multiprocessors in which the algorithm is executed. It also has applications of simulating one architecture by another.

A *star graph*, proposed in [1], is well known to have attractive properties in major metrics for an interconnection network such as the sub-logarithmic diameter, simple routing algorithm, node and edge symmetric, regular and recursive structure, and fault-tolerance properties [1], [3]. Many embedding results considering a star graph as the host graph have been intensively investigated for popular guest graphs such as ring, mesh, hypercube, and complete binary tree [2], [4], [5], [6].

Although a star graph has been evaluated as an efficient interconnection network topology, its lack of flexibility due to the exponentially increasing gap between two consecutive realizations leads to the difficulty in practical application. Thus many reserchers have been devoted to explore more practical topology in the star graph classes, and then suggested many new topologies. These approaches aim to improve the scalability in selecting the size of nodes in practical design.

As the parallel computing environment be more complex and the scale be larger, the possibility of occurrence of faulty components - nodes, edges or both - may increase. If some components fail, it is desirable that the faulty components be isolated from the rest of the network so that the guest graphs still can be embedded into it. Therefore, for the more reliable mapping scheme, a considering of fault-tolerant embedding is necessary.

The embeddability of a ring in an interconnection network is an important metrics for such applications as the trip-based multi-casting, in which an algorithm works along the containing cycle [7].

The fault-tolerant ring embedding result on the star graph has been well explored in [2], [8]. Chang *et. al* showed that it is possible to increase the embeddable ring size up to $n! - 2f_n$ in an n-dimensional star graph with $n!$ nodes provided that the sum of f_n faulty nodes and f_e faulty edges is no more than $n-3$. In case of (n, k)-star graphs, research has been mainly focused on the Hamiltonian properties [9], [10].

In this paper, we consider the ring embedding problem in a faulty (n, k)-star graph. We propose a scheme which can construct a ring of length $n!/(n-k)! - f$ in an (n, k)-star graph with at most $n-3$ faulty nodes.

The rest of this paper is organized as follows. A basic terminologies and the associated properties will be explained in Section 2. In Section 3, we analyze the basic properties of an (n, k)-star graph and propose a ring embedding scheme in the faulty (n, k)-star graphs. Finally, we conclude the paper in Section 4.

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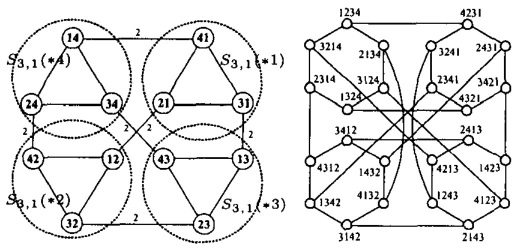

(a) (4, 2)-star graph (b) 4-star graph

Figure 1: **The $(4,2)$-star and 4-star graphs.**

2 Preliminaries

2.1 Basic Notations

An *embedding $\mathcal{F}$* of a *guest graph G* into a *host graph H* is a map of the nodes of G into the nodes of H in one-to-one fashion, accompanied by a map of the edges of G into the simple paths of H so that if $e = (x, y) \in E(G)$, then $\mathcal{F}(e)$ is a simple path of H with two end nodes $\mathcal{F}(x)$ and $\mathcal{F}(y)$.

Let $\langle n \rangle = \{1, 2, \cdots, n\}$. Then a permutation on $\langle n \rangle$ can be defined as a bijection from $\langle n \rangle$ to itself.

An n-dimensional star graph (in short *n-star*), denoted by S_n, is defined as the regular graph with the $n!$ nodes and $(n-1)n!/2$ edges, where each node is labeled by a distinct permutation of $\langle n \rangle$, and two nodes are joined by an edge along dimension d if and only if the label of one node can be obtained from the other by swapping the first and d-th symbols for any $d = 2, \cdots, n$.

An (n, k)-star graph is recently proposed by Chiang *et. al* as an attractive interconnection network topology [11]. An (n, k)-star graph is known as the generalized version of an n-star graph with scalability such that the number of nodes in the graph can be suitably adjustable by two dimensioning parameters n and k.

Definition 1 *An (n, k)-star graph, denoted by $S_{n,k}$, is specified by two integers n and k, where $1 \le k < n$. The node set of (n, k)-star graph is denoted by $\{\ p_1p_2 \cdots p_k \mid p_i \in \langle n \rangle$ and $p_i \neq p_j$ for $i \neq j\}$. An edge is connected in the following manner: a node $p_1p_2 \cdots p_i \cdots p_k$ is connected to the node (1) $p_ip_2 \cdots p_1 \cdots p_k$ by an edge of dimension i, where $2 \le i \le k$, and (2) $xp_2 \cdots p_k$ by an edge of dimension 1, where $x \in \langle n \rangle \setminus \{p_i \mid 1 \le i \le k\}$. The edges of type (1) are referred to as i-*edges, and those of type (2) are referred to as* 1-edges.

The (n, k)-star and n-star graphs are illustarted in Figure 1.

2.2 Basic Properties

An (n, k)-star graph is a regular graph of degree $n-1$ with $n!/(n-k)!$ nodes and $(n-1)n!/(2(n-k)!)$ edges. Moreover an (n, k)-star graph has a recursive structure like the star graph. This implies that an (n, k)-star graph can be decomposed into n number of $(n-1, k-1)$-star graphs as explained in the following.

Definition 2 *Let $S = (s_1s_2 \cdots s_k)_n$ be such string representation of length k that $s_i \in \langle n \rangle \cup \{*\}$ for $1 \le i \le k$. If S has exactly l number of $*$'s in dimensions except first in its representation, then S denotes $(n-k+l, l)$-sub-star decomposed from (n, k)-star graph, where $1 \le l < k$. The special symbol '$*$' denotes "don't care."*

Thus a sub-star represented by S correspons to the sub-graph of $S_{n,k}$ containg all nodes obtained from S by replacing each $*$ with the symbol in $\langle n \rangle \setminus (\bigcup_j \{x_j | x_j \neq *\})$

Definition 3 *Let's consider two $(n-k+l, l)$-sub-stars X and Y in $S_{n,k}$, where $l < k \le n$. They are defined to be* adjacent *if their string representations differ in exactly one non-$*$ position. If X and Y are adjacent, the* difference in symbol from X to Y, *denoted by difS(X, Y), is defined as the symbol of X at the position where they differ.*

For example, consider two adjacent $(4, 2)$-sub-star $X = (2**13)_7$ and $Y = (2**43)_7$ in $S_{7,5}$. For the symbol differences, *difS*$(X, Y) \neq$ *difS*(Y, X), since *difS*$(X, Y) = 1$ and *difS*$(Y, X) = 4$.

Definition 4 *Let $X = (x_1 \cdots x_j \cdots x_k)_n$ be a $(n-k+l, l)$-sub-star with $x_j = *$. The j-expansion on X, $2 \le j \le k$, is to expand X along the j-th dimension into $n-k+l$ number of $(n-k+l-1, l-1)$-sub-stars, each obtained from X by replacing x_j with a legal non-$*$ symbol. Let $D = (d_1, \cdots, d_m)$, $m < l$, be an ordered sequence of the dimensions such that $x_{d_i} = *$ for any $i = 1 \cdots m$. Then the D-expansion on X is to apply d_1-expansion, d_2-expansion, $\cdots$, d_m-expansion on that order to X. Eventually, X becomes $(n-k+l)(n-k+l-1)\cdots(n-k+l-m+1)$ number of $(n-k+l-m, l-m)$-sub-stars.*

For instance, let us consider a $(5, 2)$-sub-star $X = (21**5)_8$ in $S_{8,5}$. Then, after applying the 4-expansion on X, we get the following five number of $(4, 1)$-sub-stars: $(21*35)_8, (21*45)_8, (21*65)_8, (21*75)_8, and (21*85)_8$.

Note that the first(leftmost) dimension is also specially regarded so that we cannot apply the expansion along it since the resulting sub-stars cannot maintain the desired property any more.

Thus an (n, k)-star graph almost preserves the attractive properties of star graph. In the following, we will investigate the relationship between the (n, k)-star and n-star. The following lemma says that an n-star is a special case of an (n, k)-star graph.

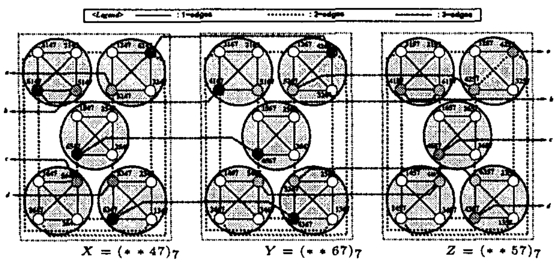

Figure 2: **An ajacency relation between two sub-stars.**

Property 1 ([11]) *An $(n, n-1)$-star graph $S_{n,n-1}$ is isomorphic to the n-star graph S_n.*

Moreover, the following lemma shows that an (n, k)-star graph possesses a special property when $k = 1$.

Lemma 1 ([8]) *An $S_{m,1}$ is isomorphic to K_m, where K_m means the complete graph of m nodes and $m(m-1)/2$ edges.* □

Now we intend to expand the notion of the ring into the super-ring consisting of super-nodes and super-edges.

Definition 5 *A sequence of $(n-k+l, l)$-sub-stars $[X_0, X_1, \cdots, X_{r-1}]$ is called as an $(n-k+l, l)$-super-ring of length r denoted by $\mathcal{R}^r_{n-k+l,l}$, where $l < k$, if the $(n-k+l, l)$-sub-star X_i is adjacent to its two neighbors $X_{(i-1) \bmod r}$ and $X_{(i+1) \bmod r}$ for any $i = 0, \cdots, r-1$.*

For example, $R = [(***3*2)_8, (***1*2)_8, (***4*2)_8, (***4*5)_8, (***3*5)_8]$ is a (6, 4)-super-ring in $S_{8,6}$.

Here, we regard the $(n-k+l, l)$-sub-stars as the corresponding super-nodes and the edges between two adjacent $(n-k+l, l)$-sub-stars as the super-edges in $\mathcal{R}^r_{n-k+l,l}$. Moreover, we further investigate the relational properties in the inter-sub-stars and intra-sub-stars.

Let $X = (x_1x_2 \cdots x_k)_n$ and $Y = (y_1y_2 \cdots y_k)_n$ be two adjacent $(n-k+l, l)$-sub-stars in $S_{n,k}$ such that $x_p = y_p = *$ for any $p = 2, \cdots, k$. Then, by applying p-expansion, we can get $n-k+l$ number of $(n-k+l-1, l-1)$-sub-stars from X and Y, respectively. Then we can observe the following properties in these sub-stars

Observation 1 *All $(n-k+l-1, l-1)$-sub-stars in X are fully connected each other within X and those in Y too in terms of adjacency. Moreover, all $(n-k+l-1, l-1)$-sub-stars in X except one, that is x, have connections to the corresponding $(n-k+l-1, l-1)$-sub-star in Y in the one-to-one fashion, and all $(n-k+l-1, l-1)$-sub-stars in Y except one, that is y, also have connections to the corresponding $(n-k+l-1, l-1)$-sub-star in X in the same manner, where $x = (x_1x_2 \cdots x_{p-1} difS(Y,X) x_{p+1} \cdots x_k)_n$ and $y = (y_1y_2 \cdots y_{p-1} difS(X,Y) y_{p+1} \cdots y_k)_n$.* □

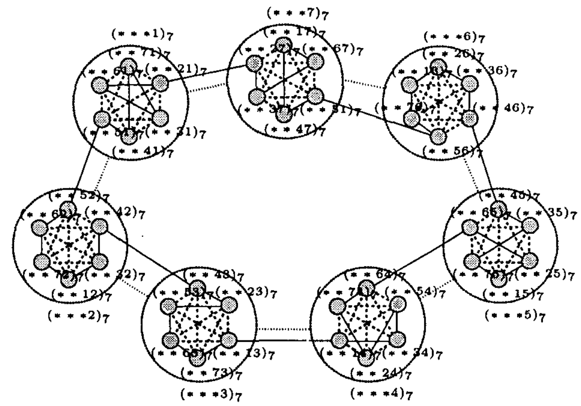

Figure 3: **An example of a super-ring construction.**

These properties are illustrated in Figure 2.

For clarity, we will omit "mod r" in the subscript that represents the element of the super-ring of length r in the rest of this paper.

3 Embedding Scheme

In this section, we present a new ring embedding scheme which can tolerate at most $n-3$ node faults in an (n, k)-star graph. Our scheme is proceeded in such a manner that first we construct an $(n-1, k-1)$-super-ring of length n from the given (n, k)-star graph, and then recursively construct a series of $(n-i-1, k-i-1)$-super-rings from $(n-i, k-i)$-super-rings until an $(n-k+1, 1)$-super-ring is obtained. In this processing, a special care must be also taken to evenly distribute faulty nodes into the resulting sub-stars. The following lemma implicitly suggests how to construct a series of super-rings.

Lemma 2 *Given an $(n-k+l, l)$-super-ring $R = [X_0, X_1, \cdots, X_{r-1}]$ in $S_{n,k}$ such that $l \geq 4$ and $n-k \geq 2$, we can construct an $(n-k+l-1, l-1)$-super-ring R' of length $r(n-k+l)$ from R.*

Proof We select a legal dimension except the first dimension, that is j, and then apply an expansion along the dimension j in each X_i, $0 \leq i \leq r-1$. Then, we can obtain $r(n-k+l)$ number of $(n-k+l-1, l-1)$-sub-stars, since $n-k+l$ number of $(n-k+l-1, l-1)$-sub-stars for each X_i. Let s be any $(n-k+l-1, l-1)$-sub-star in X_0 that has a connection to the corresponding one in X_{r-1}, then we start a construction processing of R' from

s in X_0. When we are ready to process X_i, the beginning $(n-k+l-1, l-1)$-sub-star within X_i is already known as the one connected from the last $(n-k+l-1, l-1)$-sub-star traversed in the previous X_{i-1}. As mentioned in Observation 1, all $(n-k+l-1, l-1)$-sub-stars are fully connected within X_i in terms of adjacency and there are $n-k+l-1$ (≥ 5) connections between X_i and its two neighbors X_{i-1} and X_{i+1}, respectively. Hence we can trivially traverse each X_i along the direction of R only keeping the following rules(we call this rules as *NKTR*):

- *NKR1* : Do visit all $n-k+l$ number of $(n-k+l-1, l-1)$-sub-stars within each X_i before leaving the X_i.
- *NKR2* : Do *not* arrange the $(n-k+l-1, l-1)$-sub-star in X_i with no connection to X_{i+1} in the either $(n-k+l)$-th or $(n-k+l-1)$-st traversing order within X_i (that is, arrange to any one of the first through $(n-k+l-2)$-nd).
- *NKR3* : Do *not* arrange the $(n-k+l-1, l-1)$-sub-star in X_i with no connection to X_{i-1} in the either 1-st or 2-nd traversing order within X_i (that is, arrange to any one of the third through $(n-k+l)$-th).

For preparation of smooth connection to the predetermined start $(n-k+l-1, l-1)$-sub-star s, we have to take some special attentions in processing X_{r-2}. Let t be the $(n-k+l-1, l-1)$-sub-star in X_{r-1} that has connection to the start $(n-k+l-1, l-1)$-sub-star s. Then, we have one more restriction so that the $(n-k+l-1, l-1)$-sub-star in X_{r-2} which is connected to t is also prevented from being visited in the last turn within X_{r-2}. In the worst case, three $(n-k+l-1, l-1)$-sub-stars are fixed and thus we can not align these $(n-k+l-1, l-1)$-sub-stars in the last within X_{r-1} as follows: the entry $(n-k+l-1, l-1)$-sub-star from the X_{r-3}, the $(n-k+l-1, l-1)$-sub-star that has no connection to X_{r-1}, and the $(n-k+l-1, l-1)$-sub-star that is connected to the $(n-k+l-1, l-1)$-sub-star t. Fortunately there is no problem despite of this restriction since there remains at least two $(n-k+l-1, l-1)$-sub-stars freely in X_{r-2} when $l \geq k-n+6$. By this adjustment in X_{r-2}, it can be verified to connect from t to s, thus the lemma holds. □

This recursive construction scheme is illustrated in Figure 3.

For even distribution of faulty nodes into the resulting sub-stars, it also rerquired to take an attention to select a dimension for expansion processing. The following lemma suggests the sufficient condition for $S_{n,k}$ to guarantee this partition.

Lemma 3 ([8]) *Given $S_{n,k}$, $n-k \geq 2$, with $f \leq n-3$ faulty nodes, and an integer l such that $1 \leq l < k$, we can always find a D-expansion, $|D| = l$, on $S_{n,k}$, which results in $n(n-1)\cdots(n-l+1)$ number of $(n-l, k-l)$-sub-stars and each of which contains at most $n-l-3$ faulty nodes.* □

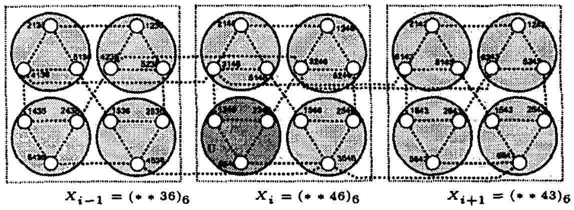

Figure 4: **An adjacency conditions for Hamiltonicity.**

We also see that the difference between two values n and k has an important role than the values n and k themselves in $S_{n,k}$ The following observation suggests the meaning of these values.

Observation 2 *The more the gap between n and k in an (n, k)-star graph, the larger complete graph is contained within the graph as a sub-graph.*

This observation implies that the (n, k)-star graph with small gap between the values n and k is less flexible than one with large gap. Thus we specially take attentions to the case of small gap between the values n and k.

Here we have to regard the special condition that confines the adjacency relation as in $S_{n,k}$.

In Figure 4, we see that it is impossible to align the special $(4, 2)$-sub-star, that is C, in X_i in processing of constructing a resulting $(4, 2)$-super-ring from the given $(5, 3)$-super-ring in any case. This implies that we can not construct an $(n-k+l-1, l-1)$-super-ring from the given $(n-k+l, l)$-super-ring in some cases. Fortunately, we also see that this is directly related to such condition that $difS(X_{i-1}, X_i) = difS(X_{i+1}, X_i)$ in the given $(n-k+l, l)$-super-ring. We also see that the visiting order of $(n-k+l-1, l-1)$-sub-stars in each $(n-k+l, l)$-sub-star in the given $(n-k+l, l)$-super-ring confines the inter-sub-star property of the resulting $(n-k+l-1, l-1)$-super-ring.

The Lemma 2 and Lemma 3 consequently imply that, after applying a series of $k-l$ number of legal expansions into $S_{n,k}$, we can obtain an $(n-k+l, l)$-super-ring of length $n(n-1)\cdots(n-k+l+1)$, each $(n-k+l, l)$-sub-star of which contains at most $n-k+l-3$ faulty nodes. For more precise analysis, we classify the possible cases as follows according to the value of $n-k$.

- *Case of $n-k=1$* : By Property 1, $S_{n,n-1}$ is equivalent to S_n. Thus it is already explored in [8]
- *Case of $n-k=2$* : This case will be considered in Subsection 3.1.

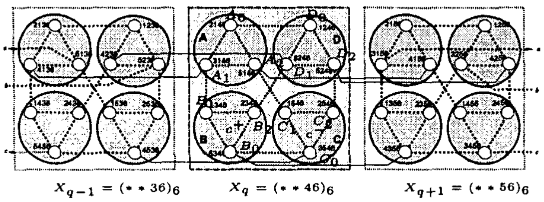

Figure 5: **A fault-free ring construction in a** $(4, 2)$-**super-ring.**

- *Case of* $n - k \geq 3$: This case will be considered in Subsection 3.2.

3.1 Case of $n - k = 2$

In this case, we eventually obtain a $(4, 2)$-super-ring by repeatedly applying $k - 2$ expansions, and each $(4, 2)$-sub-star of which contains at most one fault in $S_{n,k}$. However, for this desirable $(4, 2)$-super-ring, additional conditions must be satisfied in the previous step of super-ring construction processing. The following lemma denotes what is the pre-requisition conditios, and how a fault-free ring can be constructed from this kind of $(4, 2)$-super-rings.

Lemma 4 *Given a* $(4, 2)$-*super-ring* $R = [X_0, X_1, \cdots, X_{r-1}]$ *such that*

(1) each X_i *contains at most one faulty node,*

(2) two consecutive $(4, 2)$-*sub-stars can not contain fault simultaneously, and*

(3) $difS(X_{i-1}, X_i) \neq difS(X_{i+1}, X_i)$ *for any* $i = 0, \cdots, r - 1$,

then we can always construct a ring of length $12r - f$ *in* R.

Proof For any $k = 0, \cdots, r - 1$, we assume that X_k be a $(4, 2)$-sub-star which has one fault within it. Then, by the given condition (2), we see that any fault cannot be contained in neither X_{k-1} nor X_{k+1}. Let c^+ and c^- be the $(3, 1)$-sub-stars that have no connections to X_{k-1} and X_{k+1} (that is, have only connections to X_{k+1} and X_{k-1}) respectively. Without loss of generality, we assume that the following senarios, according what is the incomming $(3, 1)$-sub-star to X_k after traversing all $(3, 1)$-sub-stars in X_{k-1}.

- If the incomming sub-star is neither c^- nor c^+, we further classify all possible cases into the followings, according which $(3, 1)$-sub-star is faulty.
  - If the $(3, 1)$-sub-star A is faulty in Figure 5, then we can construct a path in X_k referring to one of the followings.
    * If the node A_0 is faulty, then a possible path is as follows: $A_1 - A_2 - C_1 - C_2 - C_0 -- B_0 - B_1 - B_2 - D_1 - D_0 - D_2$
    * If the node A_1 is faulty, then a possible path is as follows(in this case, it is necessary additionary to adjust the incomming sub-star into C by re-routing in the previous $(4, 2)$-sub-star X_{k-1}): $C_0 - C_2 - C_1 - A_2 - A_0 -- D_0 - D_2 - D_1 - B_2 - B_1 - B_0$
    * If the node A_2 is faulty, then a possible path is as follows: $A_1 - A_0 - D_0 - D_1 - B_2 -- B_1 - B_0 - C_0 - C_1 - C_2 - D_2$
  - If the $(3, 1)$-sub-star B is faulty in Figure 5, then it is also possible to construct a path in X_k by referring to one of the followings.
    * If the node B_0 is faulty, then a possible path is as follows: $A_1 - B_1 - B_2 - D_1 - D_0 -- A_0 - A_2 - C_1 - C_0 - C_2 - D_2$
    * If the node B_1 is faulty, then a possible path is as follows: $A_1 - A_0 - A_2 - C_1 - C_0 -- C_2 - D_2 - D_0 - D_1 - B_2 - B_0$
    * If the node B_2 is faulty, then a possible path is as follows: $A_1 - B_1 - B_0 - C_0 - C_1 -- C_2 - D_2 - D_1 - D_0 - A_0 - A_2$
  - If the $(3, 1)$-sub-star C is faulty in Figure 5, then we can construct a path in X_k referring to one of the followings.
    * If the node C_0 is faulty, then a possible path is as follows: $A_1 - A_0 - A_2 - C_1 - C_2 -- D_2 - D_0 - D_1 - B_2 - B_1 - B_0$
    * If the node C_1 is faulty, then a possible path is as follows: $A_1 - B_1 - B_2 - B_0 - C_0 -- C_2 - D_2 - D_1 - D_0 - A_0 - A_2$
    * If the node C_2 is faulty, then a possible path is as follows: $A_1 - A_0 - D_0 - D_2 - D_1 -- B_2 - B_1 - B_0 - C_0 - C_1 - A_2$
  - If the $(3, 1)$-sub-star D is faulty in Figure 5, then it is also possible to construct a path in X_k by referring to one of the followings.
    * If the node D_0 is faulty, then a possible path is as follows: $A_1 - A_0 - A_2 - C_1 - C_2 -- C_0 - B_0 - B_1 - B_2 - D_1 - D_2$
    * If the node D_1 is faulty, then a possible path is as follows: $A_1 - B_1 - B_2 - B_0 - C_0 -- C_1 - C_2 - D_2 - D_0 - A_0 - A_2$
    * If the node D_2 is faulty, then a possible path is as follows: $A_1 - A_0 - D_0 - D_1 - B_2 -- B_1 - B_0 - C_0 - C_2 - C_1 - A_2$
- If the incomming sub-star corresponds to c^-, we further classify the possible cases into the followings, according as which $(3, 1)$-sub-star is faulty.
  - If the $(3, 1)$-sub-star A is faulty in Figure 5, then it is also possible to construct a path in X_k by referring to one of the followings.
    * If the node A_0 is faulty, then a possible path is as follows: $C_0 - C_1 - C_2 - D_2 - D_0 -- D_1 - B_2 - B_0 - B_1 - A_1 - A_2$

* If the node A_1 is faulty, then a possible path is as follows: $C_0 - C_2 - C_1 - A_2 - A_0 -- D_0 - D_2 - D_1 - B_2 - B_1 - B_0$
* If the node A_2 is faulty, then a possible path is as follows: $C_0 - C_1 - C_2 - D_2 - D_1 -- D_0 - A_0 - A_1 - B_1 - B_2 - B_0$

- If the $(3,1)$-sub-star B is faulty in Figure 5, then it is also possible to construct a path in X_k by referring to one of the followings.
  * If the node B_0 is faulty, then a possible path is as follows: $C_0 - C_2 - C_1 - A_2 - A_0 -- A_1 - B_1 - B_2 - D_1 - D_0 - D_2$
  * If the node B_1 is faulty, then a possible path is as follows: $C_0 - C_2 - C_1 - A_2 - A_1 -- A_0 - D_0 - D_2 - D_1 - B_2 - B_0$
  * If the node B_2 is faulty, then a possible path is as follows: $C_0 - B_0 - B_1 - A_1 - A_0 -- D_0 - D_1 - D_2 - C_2 - C_1 - A_2$
- If the $(3,1)$-sub-star C is faulty in Figure 5, then we can construct a path in X_k referring to one of the followings.
  * If the node C_0 is faulty, then a possible path is as follows(in this case, it is necessary additionary to adjust the incomming sub-star into A by re-routing in the previous $(4,2)$-sub-star X_{k-1}): $A_1 - A_0 - A_2 - C_1 - C_2 -- D_2 - D_0 - D_1 - B_2 - B_1 - B_0$
  * If the node C_1 is faulty, then a possible path is as follows: $C_0 - C_2 - D_2 - D_1 - D_0 -- A_0 - A_2 - A_1 - B_1 - B_2 - B_0$
  * If the node C_2 is faulty, then a possible path is as follows: $C_0 - C_1 - A_2 - A_1 - A_0 -- D_0 - D_2 - D_1 - B_2 - B_1 - B_0$
- If the $(3,1)$-sub-star D is faulty in Figure 5, then we can construct a path in X_k referring to one of the followings.
  * If the node D_0 is faulty, then a possible path is as follows: $C_0 - C_1 - C_2 - D_2 - D_1 -- B_2 - B_0 - B_1 - A_1 - A_0 - A_2$
  * If the node D_1 is faulty, then a possible path is as follows: $C_0 - C_1 - C_2 - D_2 - D_0 -- A_0 - A_2 - A_1 - B_1 - B_2 - B_0$
  * If the node D_2 is faulty, then a possible path is as follows: $C_0 - C_2 - C_1 - A_2 - A_1 -- A_0 - D_0 - D_1 - B_2 - B_1 - B_0$

In this analysis, there is no problem in two special cases which need re-route in the previous $(4,2)$-sub-star X_{k-1} because the previous sub-star does not contain any fault. Thus the lemma holds. □

Lemma 4 says that we have to take a special attention when we prepare a $(4,2)$-super-ring from the given $(5,3)$-super-ring. This consideration leads to another conditions in $(5,3)$-super-ring as follows.

Lemma 5 *Given a $(5,3)$-super-ring $R = [X_0, X_1, \cdots, X_{r-1}]$ for any $i = 0, \cdots, r-1$, we can always construct a $(4,2)$-super-ring $R' = [Y_0, Y_1, \cdots, Y_{5r-1}]$ from R such that $difS(Y_{j-1}, Y_j) \neq difS(Y_{j+1}, Y_j)$ for any $j = 0, \cdots, 5r-1$.*

Proof We will prove by constructive method. First, we select a dimension j for an expansion so that each of the resulting $(4,2)$-sub-stars contains at most one faulty node. We readily see that at least one $(5,3)$-sub-star contains no faults in R since the number of total faults is sufficiently small. Hence we assume that X_0 is fault-free, otherwise we can re-label the given R to position a fault-free $(5,3)$-sub-star at the first. Starting from the initial $(5,3)$-sub-star X_1, we traverse all $(5,3)$-sub-stars in R one by one along the super-edge and finally extend to the $(4,2)$-super-ring that includes all $(4,2)$-sub-stars expanded once more from R.

Let S and E be the start and end $(4,2)$-sub-stars in the resulting $(4,2)$-super-ring. We will set E to the $(4,2)$-sub-star in X_0 that is not connected to any one in X_{r-1}, which results in the last one visited in X_0. This ensure that, at the last step, the start and end $(4,2)$-sub-stars are distinct in X_0. Thus the first $(4,2)$-sub-star to be traversed within X_1 is automatically decided as the one that has connection to it, which will be set to S. Thus, at the time to process each X_i, the starting $(4,2)$-sub-star of X_i is already known as the $(4,2)$-sub-star that has a connection to the last $(4,2)$-sub-star traversed within X_{i-1}. From the given start $(4,2)$-sub-star, we have to decide the traversing sequence of five $(4,2)$-sub-stars within X_i to satisfy the given conditions.

Here we assume the following scenario. Initially, we obviously see that the 4-sub-stars within X_0 can be assigned to satisfy the given conditions because it contains no faults. In the k-th step, $1 \leq k \leq r-1$, it is assumed that the processing of X_{k-1} has been completed to satisfy the given conditions, and then we are ready to decide the traversing sequence of the five $(4,2)$-sub-stars in X_k to satisfy the conditions. For this purpose, we will consider all possibilities within X_k. Let s be the first $(4,2)$-sub-star visited within X_k, which may be already fixed because of its dependency on the last $(4,2)$-sub-star visited within X_{k-1}. Let c^+ and c^- be the $(4,2)$-sub-stars that have no connections to X_{k-1} and X_{k+1}(that is, have only connections to X_{k+1} and X_{k-1}), respectively.

As the worst case, we assume that two $(4,2)$-sub-stars contain faulty nodes within X_k. Then we regard the following two cases whether the $(4,2)$-sub-star s may be faulty or not.

- If the start $(4,2)$-sub-star s is faulty, then the other one faulty $(4,2)$-sub-star will be assigned to traverse in the third or any one of third and fourth whether the fault occurs in the $(4,2)$-sub-star c^- or not without violating the given conditions within X_k.

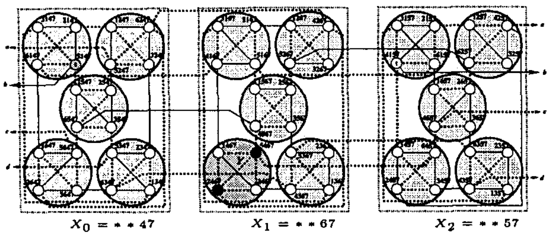

Figure 6: **An example of ring construction.**

- If s is non-faulty, then remaining two faulty $(4,2)$-sub-stars can be always assigned to traverse in the second and fourth by keeping such principles that *"the faulty* $(4,2)$*-sub-stars occurred in* c^- *and* c^+ *are certainly assigned to traverse in the second and fourth respectively, and the other(s) in the second or fourth arbitrarily if it is not already assigned within* X_k*."* Obviously the given conditions are satisfied in this assignment.

In such case that only one $(4,2)$-sub-star contains faulty node within X_k, we can readily assign all $(4,2)$-sub-stars to satisfy the given conditions, but details will be omitted. At the last step, when we reach to X_0, we can construct such path from the start $(4,2)$-sub-star in X_0 to E so that all five $(4,2)$-sub-stars in X_0 are traversed without violating the *NKTR*, because X_0 is totally fault-free and the five $(4,2)$-sub-stars within X_0 are fully connected. Consequently we can construct a $(4,2)$-super-ring satisfying the given conditions by connecting the $(4,2)$-sub-star E to S. □

By summarizing Lemma 2, Lemma 5, and Lemma 4, we can see the following theorem true.

Theorem 1 *If an* $S_{n,k}$ *with* f *faulty nodes satisfies such condition that* $f \leq n-3$ *when* $n-k=2$*, then we can always find a fault-free ring of length* $n!/(n-k)! - f$ *in the* $S_{n,k}$*.* □

3.2 Case of $n-k \geq 3$

By Observation 2, we see that it will be more flexible in constructing paths in sub-stars. The following lemma denotes what is the pre-requisition conditios, and how a fault-free ring can be constructed from this kind of super-rings.

Lemma 6 *Given an* $(n-k+2,2)$*-super-ring* $R = [X_0, X_1, \cdots, X_{r-1}]$ *such that*

(1) each X_i *contains at most* $n-k-1$ *faulty nodes, and*

(2) $difS(X_{i-1}, X_i) \neq difS(X_{i+1}, X_i)$ *for any* $i = 0, \cdots, r-1$,

then we can always construct a ring of length $r(n-k+1)(n-k+2) - f$ *in R.*

Proof After applying a legal j-expansion, we can obtain $n-k+2$ number of $(n-k+1,1)$-sub-stars for each $(n-k+2,2)$-sub-star X_i in R. Then, in such a manner as shown in Lemma 2, we intend to construct fault-free path by keeping the *NKTR* rules. For more analysis, we can classify the possible cases as follows.

1. Let us consider the case that faults are distributed into at least two $(n-k+1,1)$-sub-stars. As explained in Observation 1 and Lemma 1, we already see the following two properties in the relationship of inter-$(n-k+1,1)$-sub-star and intra-$(n-k+1,1)$-sub-star: (1) the $(n-k+1,1)$-sub-stars within X_i are completely connected each other, and (2) the nodes within each $(n-k+1,1)$-sub-star are also fully connected each other. Thus, under any fault distribution, there are sufficient possibilities in connection for constructing a path because the number of faults contained within each $(n-k+1,1)$-sub-star is sufficiently small than those of faults-free nodes. So we can readily obtain the resulting ring by continuosly expanding of the fault-free path into the next $(n-k+2,2)$-sub-star in R by keeping the *NKTR* rules.

2. Let us consider the case that all faults are concentrated into only one $(n-k+1,1)$-sub-star(that is, Y in Figure 6). In this case, only two non-faulty nodes are remained in Y. Thus the pre-positioned and post-positioned $(n-k+1,1)$-sub-stars in traversing are automatically fixed as the two fault-free nodes which have the remaining connection to outside of Y. Because of $n-k \geq 3$, there always exists at least one position in order into which Y may be aligned without violating the *NKTR* rules. Thus, by fixing the traversing order of Y into third, the second and fourth order of tarversing sequence are also fixed in X_i. Then, we see that at least one of the remaining $(n-k+1,1)$-sub-stars which are not assigned their traversing order yet evidently exists. So one of these can be assigned as the first order position in traversing in X_i. This ordering is important since this assignment does not violate any rule of the *NKTR*. However, in some cases, this assignment may be cause a re-routing in the previous $(n-k+2,2)$-sub-star of R. However, there is no problem since the previous $(n-k+2,2)$-sub-star can not contain any fault within it because there is no fault remained in the other part than Y. Thus it is possible to connect all fault-free nodes into the reselting ring by keeping the *NKTR* rules.

In any case, there is no problem to construct a path consisted of all non-faulty nodes in the given R. Thus the lemma holds. □

This lemma does not require any other special condition any more. Thus we can reaidy prepare a $(n-k+2,2)$-

super-ring by using the prebiously described Lemma 2.

Now we conclude this subsection to the following theorem.

Theorem 2 *If an $S_{n,k}$ with f faulty nodes satisfies such condition that $f \leq n-3$ when $n-k \geq 2$, then we can always find a fault-free ring of length $n!/(n-k)! - f$ in the $S_{n,k}$.* □

3.3 Embedding Algorithm

By summarizing our discussion, we present the following algorithm which can embed a ring of length $n!/(n-k)!-f$ in an (n,k)-star graph with f faulty nodes provided that $n-k \geq 2$.

Algorithm *Ring-in-FNKS()*

1. Find a sequence of dimensions $D = (d_k, d_{k-1}, \cdots, d_2)$ for expansions as in Lemma 3.
2. Apply a d_k-expansion on $S_{n,k}$, and construct an $(n-1, k-1)$-super-ring $R_{n-1,k-1}$ of length n from $S_{n,k}$.
3. for $j := k-1$ down-to 4 do
   Apply a j-expansion on $R_{n-k+j,j}$ and
   then use Lemma 2 to construct an $(n-k+j-1, j-1)$-super-ring $R_{n-k+j-1,j-1}$ from $R_{n-k+j,j}$
4. Apply a d_3-expansion on $R_{n-k+3,3}$ and construct an $(n-k+2, 2)$-super-ring $R_{n-k+2,2}$ from $R_{n-k+3,3}$ by using either Lemma 5 if $n-k=2$, or Lemma 2 otherwise.
5. Apply a d_2-expansion on $R_{n-k+2,2}$ and directly construct a fault-free ring from $R_{n-k+2,2}$ by using either Lemma 4 if $n-k=2$, or Lemma 6 otherwise.

The preceding lemmas imply the correctness of the Algorithm *Ring-in-FNKS*.

4 Conclusion

In this paper, we considered the ring embedding problem in the faulty (n,k)-star graphs. We presented a new fault-tolerant ring embedding scheme that utilizes the recursive structure of the graph itself. We also used the method which can evenly distribute faults into the resulting substars. These strategies lead to embedding a ring of length $n!/(n-k)!-f$ into an (n,k)-star with f faulty nodes when $f \leq n-3$. Our result can be evaluated as the worst case optimal with respect that the maximum number of allowable fault nodes is trivially bounded to two less than the minimum degree of the graph.

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